

Report
Of
Advanced Robotics Using AI And IoT Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student:2410030182

Name of Student: ANURAG BETAL

Batch:

2026-27

CANDIDATE'S DECLARATION

I, Anurag Betal, Roll No. 2410030182, do hereby declare that the internship report titled **“Advanced Robotics Using AI And IoT Virtual Internship”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Anurag Betal

Roll No: 2410030182

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name: Anurag Betal

Roll No: 2410030182

INTERSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
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NATIONAL
INTERNSHIP
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EduSkills®
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Certificate of Virtual Internship

This is to certify that

Anurag Betal

IILM University, Greater Noida

has successfully completed the 8-weeks

Advanced Robotics Using AI And IoT Virtual Internship

During June - August 2026

Supported by



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
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Certificate ID : 4733adbcaa925836d273
Student ID: STU6997425f4f04a1771520607



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	i
2.	Acknowledgment	ii
3.	Internship Completion Certificate	iii
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof must be attached	1 2 3 4 5 6-7 8-10 10-12 12-13 13-14
5.	Bibliography/References	15

CHAPTER 4:PROJECT DESCRIPTION

4.1 Introduction

Advanced Robotics is an emerging field that combines robotics, Artificial Intelligence (AI), and the Internet of Things (IoT) to develop intelligent and autonomous systems. The integration of these technologies enables robots to sense their environment, process information, make decisions, communicate with other devices, and perform tasks with minimal human intervention.

The **Advanced Robotics Using AI and IoT Virtual Internship** provides practical exposure to the concepts, technologies, and applications involved in modern robotic systems. During the internship, various aspects of robotics were explored, including sensors, actuators, microcontrollers, robot control, communication technologies, Artificial Intelligence, and IoT-based monitoring. AI techniques help robots analyze sensor data and make intelligent decisions, while IoT enables communication between robotic devices, cloud platforms, and users over a network.

The project focuses on understanding how AI and IoT can be integrated with robotics to create smart, efficient, and connected robotic systems. Sensors are used to collect information from the surroundings, which is processed by the robotic controller. Based on the processed information, the robot can perform appropriate actions through motors or other actuators. IoT connectivity further allows the collected data and system status to be monitored remotely.

This internship helped in developing a practical understanding of robotic system design, automation, intelligent decision-making, and IoT-based communication. It also provided valuable knowledge about the role of advanced technologies in applications such as industrial automation, smart environments, healthcare, agriculture, surveillance, and autonomous systems. Overall, the internship served as a foundation for understanding and developing next-generation intelligent robotic solutions.

4.2 Organization Profile

EduSkills is a non-profit organization focused on providing students with industry-oriented learning opportunities and helping bridge the gap between academic education and the rapidly changing requirements of the technology industry. The organization works toward developing skilled and employable graduates by providing access to technical training, virtual internships, certifications, and practical learning experiences.

EduSkills focuses on emerging technologies and industry-relevant domains such as **Artificial Intelligence, Machine Learning, Internet of Things (IoT), Robotics, Cloud Computing, Cybersecurity, Data Science, and Networking**. Through its various learning and internship initiatives, students get an opportunity to understand modern technologies and apply their theoretical knowledge to practical and real-world problems.

The organization collaborates with **industry and technology partners** to provide students with structured learning programs and virtual internship opportunities. These programs are designed to improve technical knowledge, practical skills, problem-solving abilities, and awareness of current industry practices.

The **Advanced Robotics Using AI and IoT Virtual Internship** is one such learning opportunity that provides exposure to the integration of robotics with Artificial Intelligence and Internet of Things technologies. During the internship, students learn about robotic systems, sensors, actuators, microcontrollers, automation, AI-based decision-making, and IoT-based communication and monitoring.

4.3 Problem Statement

Traditional robotic systems often depend on predefined instructions and limited sensing capabilities, which restrict their ability to respond intelligently to changing environmental conditions. Many conventional robots also operate independently without continuous communication, remote monitoring, or intelligent decision-making capabilities.

There is a need to develop robotic systems that can **sense their surroundings, process real-time information, make intelligent decisions, and communicate with users or other devices**. Integrating **Artificial Intelligence (AI)** with **Internet of Things (IoT)** can overcome these limitations by enabling robots to analyze sensor data, make autonomous decisions, and transmit information through connected networks.

The problem addressed in this project is to understand and develop an **AI- and IoT-enabled robotic system** that can collect information from its environment using sensors, process the collected data intelligently, perform appropriate actions through actuators, and provide remote monitoring and communication capabilities.

The project therefore focuses on exploring how **Robotics, AI, and IoT** can be integrated to create smarter, more autonomous, connected, and efficient robotic systems suitable for real-world applications such as automation, surveillance, healthcare, agriculture, and smart environments.

4.4 Project Objectives

The main objective of the **Advanced Robotics Using AI and IoT Virtual Internship** is to understand the integration of Artificial Intelligence, Internet of Things, and Robotics for developing intelligent and automated robotic systems.

The major objectives of the project are:

1. **To understand the fundamentals of robotics** and learn about the basic components of robotic systems, including sensors, actuators, motors, and controllers.
2. **To explore the application of Artificial Intelligence in robotics** for intelligent decision-making, automation, and autonomous operation.
3. **To understand IoT-based communication** and learn how robotic systems can exchange data with connected devices and remote platforms.
4. **To work with sensors and actuators** for collecting environmental information and controlling the physical movement or actions of a robotic system.
5. **To develop an understanding of real-time data processing**, where sensor data can be collected, analyzed, and used to make appropriate decisions.
6. **To study automation and autonomous robotic operations**, reducing the need for continuous human intervention.
7. **To understand remote monitoring and control** of robotic systems using IoT technologies.
8. **To integrate AI, IoT, and robotics concepts** to understand the development of smart and connected robotic applications.
9. **To develop practical and problem-solving skills** by applying theoretical concepts to robotics-based applications.
10. **To gain industry-oriented technical knowledge** in emerging technologies such as Artificial Intelligence, IoT, robotics, automation, and embedded systems.

4.5 Scope of the Project

The scope of the **Advanced Robotics Using AI and IoT Virtual Internship** is to understand the design, operation, and application of intelligent robotic systems by integrating **Artificial Intelligence (AI), Internet of Things (IoT), robotics, sensors, actuators, and automation technologies**. The project provides a foundation for understanding how connected robotic systems can collect information, process data, make decisions, and perform tasks with reduced human intervention.

The major areas covered within the scope of the project are:

1. **Robotics Fundamentals**
Understanding the basic concepts of robotics, including robot components, movement, control systems, sensors, actuators, and robotic mechanisms.
2. **Artificial Intelligence Integration**
Exploring how AI can be used in robotics for decision-making, object recognition, environmental analysis, automation, and autonomous operation.
3. **IoT Connectivity**
Understanding how IoT technologies can connect robotic systems to networks, enabling data communication, remote monitoring, and control.
4. **Sensors and Data Collection**
Studying different sensors used in robotic systems to collect information such as distance, temperature, light, motion, and environmental conditions.
5. **Real-Time Monitoring**
Exploring methods for transmitting and monitoring robotic data through connected devices or cloud-based platforms.
6. **Automation and Autonomous Systems**
Understanding how AI and sensor-based decision-making can reduce human intervention and improve the efficiency of robotic operations.
7. **Practical Applications**
The concepts learned can be applied to areas such as industrial automation, smart agriculture, healthcare, surveillance, logistics, smart homes, and autonomous vehicles.
8. **Future Development**
The project provides a foundation for developing more advanced robotic systems using technologies such as machine learning, computer vision, cloud computing, edge computing, and advanced IoT platforms.

4.6 Technologies and Tools Used

The **Advanced Robotics Using AI and IoT Virtual Internship** involves a combination of robotics, Artificial Intelligence, Internet of Things, embedded systems, and programming technologies. These technologies help in understanding how intelligent robotic systems can sense their environment, process information, communicate with other devices, and perform automated actions.

1. Robotics

Robotics is the core technology used for understanding the design and operation of automated machines. It includes concepts such as robot movement, control mechanisms, sensors, actuators, motors, and automation.

2. Artificial Intelligence (AI)

Artificial Intelligence is used to provide intelligent decision-making capabilities to robotic systems. AI can help robots analyze collected information, identify patterns, recognize objects, and perform tasks autonomously.

3. Internet of Things (IoT)

IoT enables robotic systems to communicate with connected devices and networks. It can be used for collecting, transmitting, and monitoring data remotely.

4. Sensors

Sensors are used to collect information from the surrounding environment. Depending on the application, sensors can measure parameters such as distance, temperature, humidity, light, motion, and obstacles.

5. Microcontrollers

Microcontrollers such as **Arduino, ESP8266, or ESP32** can be used as the control unit of a robotic system. They receive sensor data, process inputs, and control motors or other components.

6. Actuators and Motors

Motors and actuators convert electrical control signals into physical movement. **DC motors and servo motors** are commonly used for controlling robotic movement and mechanical operations.

7. Embedded Systems

Embedded systems combine hardware and software to perform dedicated functions within a robotic device. They provide the interface between sensors, controllers, communication modules, and actuators.

8. Programming

Programming is required to control the robotic system and implement its logic. Languages and environments such as **C/C++**, **Python**, and **Arduino IDE** can be used depending on the application and hardware platform.

9. Communication Technologies

Communication technologies such as **Wi-Fi**, **Bluetooth**, **MQTT**, and **HTTP** can be used to exchange information between robotic systems, IoT devices, servers, and monitoring applications.

10. Cloud and IoT Platforms

Cloud or IoT platforms can be used to store, visualize, and monitor data received from connected robotic devices. They can also support remote monitoring and control.

11. Simulation and Development Tools

Simulation and development environments can be used to design, test, and analyze robotic systems before implementing them on physical hardware. These tools help reduce development time and identify errors during the design stage.

4.7 System Architecture

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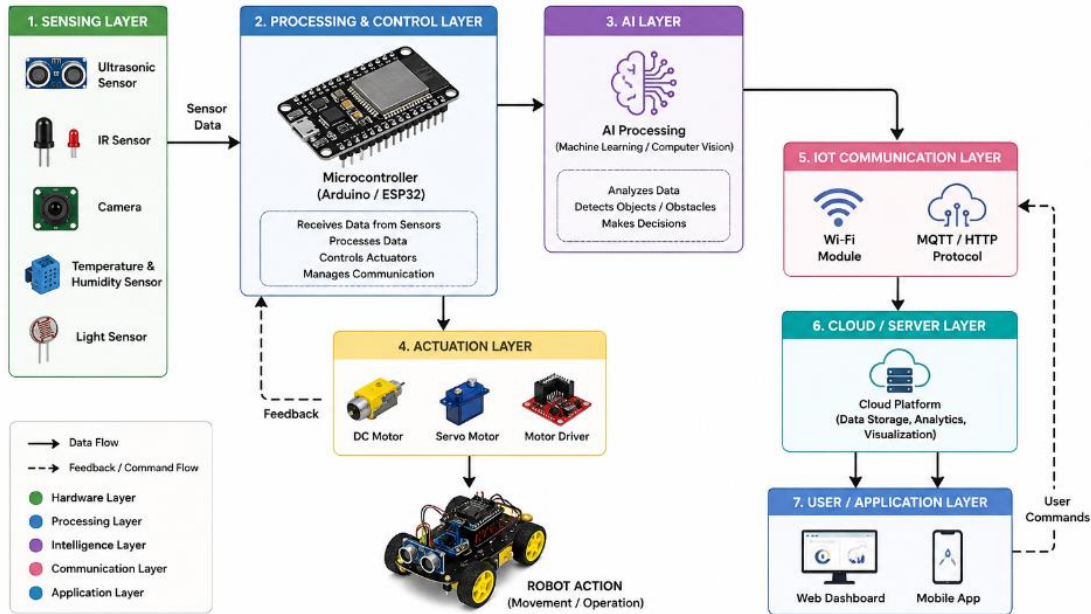
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SYSTEM ARCHITECTURE

Advanced Robotics Using AI and IoT



4.8 Methodology

The methodology of the **Advanced Robotics Using AI and IoT Virtual Internship** follows a systematic approach to understand the development and working of an intelligent, connected robotic system. The process combines robotics, sensors, Artificial Intelligence, IoT communication, data processing, and automated control.

1. Requirement Analysis

The first step is to identify the requirements of the robotic system, including the tasks to be performed, environmental conditions, required sensors, processing capabilities, communication requirements, and expected output.

2. System Design

The overall robotic system is designed by dividing it into different components such as the **sensing unit, processing unit, AI unit, communication unit, and actuation unit**. The interaction between these components is planned to ensure proper data flow and control.

3. Sensor Data Collection

Sensors are used to collect real-time information from the robot's surroundings. Depending on the application, sensors may detect distance, obstacles, temperature, humidity, light, motion, or visual information.

The collected sensor data is provided to the controller for further processing.

4. Data Processing

A microcontroller or processing unit such as **Arduino or ESP32** receives the sensor information. The data is processed to remove unnecessary information and convert the sensor readings into useful inputs for decision-making.

5. AI-Based Decision Making

Artificial Intelligence techniques are used to analyze the available data and support intelligent decision-making. Based on the input received from sensors or other sources, the system determines the appropriate action to be performed by the robot.

For example, the system can identify an obstacle and decide whether the robot should stop, turn, or change its direction.

6. Robot Control and Actuation

The decision generated by the processing system is converted into physical actions using actuators. **DC motors, servo motors, and motor drivers** can be controlled according to the required operation of the robot.

7. IoT Integration

IoT connectivity is implemented to allow the robotic system to communicate with external devices or platforms. Technologies such as **Wi-Fi, MQTT, or HTTP** can be used to transmit sensor data and system information.

8. Cloud-Based Monitoring

The data generated by the robotic system can be transmitted to a cloud or IoT platform. The information can then be stored, visualized, and monitored remotely using a web dashboard or mobile application.

9. Testing and Evaluation

The complete system is tested to verify the functioning of sensors, AI-based decision-making, communication, and robot movement. Different inputs and operating conditions are considered to identify errors and improve system performance.

10. Performance Analysis

Finally, the performance of the system is analyzed based on parameters such as **accuracy, response time, reliability, communication efficiency, and autonomous operation**. The results are used to determine whether the system meets its intended objectives.

4.9 Expected Outcomes

The **Advanced Robotics Using AI and IoT Virtual Internship** is expected to provide practical and technical knowledge about the integration of Robotics, Artificial Intelligence, and Internet of Things technologies. The major expected outcomes of the project are:

1. **Understanding of Robotics**
Gain a clear understanding of robotic systems, their components, working principles, sensors, actuators, motors, and control mechanisms.
2. **Knowledge of Artificial Intelligence**
Understand how AI and machine learning techniques can be applied to robotic systems for intelligent decision-making and autonomous operation.
3. **Understanding of IoT Integration**
Learn how IoT technologies can connect robotic systems to networks and enable communication, remote monitoring, and data exchange.
4. **Sensor and Actuator Integration**
Develop an understanding of how sensors collect environmental information and how actuators such as DC motors and servo motors perform physical actions.
5. **Real-Time Data Processing**
Understand how sensor data can be collected, processed, analyzed, and used to generate appropriate responses from the robotic system.
6. **Autonomous Operation**
Understand the basic principles of autonomous robotics, where robots can respond to environmental conditions with minimal human intervention.
7. **Remote Monitoring and Control**
Gain knowledge of monitoring robotic systems remotely through IoT networks, cloud platforms, web dashboards, or mobile applications.

8. **Practical Technical Skills**

Develop practical knowledge of microcontrollers, programming, sensors, communication protocols, and robotic control systems.

9. **Problem-Solving Skills**

Improve the ability to analyze technical problems, design solutions, troubleshoot systems, and evaluate robotic system performance.

10. **Industry-Oriented Knowledge**

Gain exposure to emerging technologies and understand their applications in areas such as industrial automation, smart agriculture, healthcare, surveillance, logistics, and smart environments.

4.10 Certificates of completion

The internship was successfully completed, and a Certificate of Internship was issued by EduSkills Academy on August 08, 2026 (Certificate ID: 4733adbcaa925836d273), confirming completion of the 8-week **Advanced Robotics Using AI and IoT Virtual Internship**. The internship provided practical exposure to **robotics, Artificial Intelligence (AI), Internet of Things (IoT), sensors, actuators, microcontrollers, automation, robotic control systems, and AI-based decision-making**. It also helped in understanding the integration of AI and IoT technologies with robotic systems for intelligent and autonomous applications. A copy of the certificate is attached in this report under "**Internship Completion Certificate**" (page iii).

Proof of Enrollment

New Domain Course Enrollment Confirmation External Inbox x

 **EduSkills Foundation** <noreply@eduskillsfoundation.org>
to me ▾ Tue, Jul 21, 6:35 PM ☆

AICTE-EduSkills Internship Registration Confirmation

Dear Anurag Betal,

You have successfully applied for the AICTE - **EduSkills** Virtual Internship in **Advanced Robotics Using AI And IoT VIRTUAL INTERNSHIP**.

No profile URL required for this domain

Your application has been sent to your institute for approval. Please contact your faculty for further process.

Faculty Name: Monika Kumari
Email ID: monika.kumari@iilm.edu
Mobile Number: 7006148254

If you are still having any issues, you may drop a mail to internship@eduskillsfoundation.org OR contact us on 8093254914 (Mon-Sat between 10.00am to 05.00pm)

Note: This is a system-generated e-mail. Please do NOT reply to it.

Thank you and have a great day!

Best Regards,
EduSkills Team

Proof of Offer letter



Ref No: C17Y2026M081517530

Internship Offer Letter

Student Name : Anurag Betal
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Advanced Robotics Using AI And IoT
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU6997425f4f04a1771520607

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

CHAPTER 5: Bibliography/References

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